

Complete 90-Day LeetCode Roadmap: 300+ Problems to Interview Mastery

Executive Summary

This comprehensive 90-day intensive program is designed to take you from beginner to interview-ready level by solving 300+ carefully curated LeetCode problems. The program requires **4-5 hours of daily commitment** and follows a strategic progression through essential algorithms and data structures using Java.

Program Overview

- **Duration:** 90 days (13 weeks)
- **Total Problems:** 340+ curated problems
- **Daily Commitment:** 4-5 hours of focused study
- **Target Outcome:** Interview readiness at top tech companies
- **Programming Language:** Java (with optimization tips)

Phase-by-Phase Roadmap

Phase 1: Foundation Building (Weeks 1-4)

Days 1-28 | Target: 100 Problems

Week 1: Arrays & Strings Foundation (25 problems)

Daily Target: 3-4 problems (20 Easy + 5 Medium)

Core Problems:

- Two Sum (Easy)
- Remove Duplicates from Sorted Array (Easy)
- Maximum Subarray (Easy)
- Product of Array Except Self (Medium)
- Valid Palindrome (Easy)
- Reverse String (Easy)
- Longest Common Prefix (Easy)

Key Java Concepts: ArrayList operations, StringBuilder, character arrays, Arrays.sort()

Week 2: Hash Tables & Two Pointers (25 problems)

Daily Target: 3-4 problems (15 Easy + 10 Medium)

Core Problems:

- Two Sum II - Input array is sorted (Easy)
- 3Sum (Medium)
- Container With Most Water (Medium)
- Group Anagrams (Medium)
- Longest Consecutive Sequence (Medium)
- Top K Frequent Elements (Medium)

Key Java Concepts: HashMap operations, HashSet, keySet(), values(), getOrDefault()

Week 3: Sliding Window & Binary Search (25 problems)

Daily Target: 3-4 problems (10 Easy + 15 Medium)

Core Problems:

- Longest Substring Without Repeating Characters (Medium)
- Minimum Window Substring (Hard)
- Search in Rotated Sorted Array (Medium)
- Find First and Last Position of Element (Medium)
- Binary Search (Easy)
- Find Peak Element (Medium)

Key Java Concepts: Deque, Collections.binarySearch(), custom comparators

Week 4: Linked Lists & Stacks/Queues (25 problems)

Daily Target: 3-4 problems (5 Easy + 20 Medium)

Core Problems:

- Reverse Linked List (Easy)
- Linked List Cycle (Easy)
- Merge Two Sorted Lists (Easy)
- Add Two Numbers (Medium)
- Remove Nth Node From End of List (Medium)
- Valid Parentheses (Easy)
- Min Stack (Easy)
- Daily Temperatures (Medium)

Key Java Concepts: ListNode class, dummy nodes, Stack<Integer>, Queue<Integer>

Phase 2: Advanced Data Structures (Weeks 5-8)

Days 29-56 | Target: 100 More Problems

Week 5: Trees & Binary Search Trees (25 problems)

Daily Target: 3-4 problems (5 Easy + 20 Medium)

Core Problems:

- Binary Tree Inorder Traversal (Easy)
- Maximum Depth of Binary Tree (Easy)
- Validate Binary Search Tree (Medium)
- Lowest Common Ancestor of a Binary Search Tree (Easy)
- Binary Tree Level Order Traversal (Medium)
- Path Sum (Easy)

Key Java Concepts: TreeNode recursion, null checks, inorder/preorder/postorder traversals

Week 6: DFS/BFS & Tree Traversals (25 problems)

Daily Target: 3-4 problems (5 Easy + 15 Medium + 5 Hard)

Core Problems:

- Number of Islands (Medium)
- Binary Tree Right Side View (Medium)
- Word Ladder (Hard)
- Clone Graph (Medium)
- Course Schedule (Medium)
- Binary Tree Zigzag Level Order Traversal (Medium)

Key Java Concepts: Queue<TreeNode>, level-order traversal, Set<TreeNode> for visited nodes

Week 7: Dynamic Programming Basics (25 problems)

Daily Target: 3-4 problems (3 Easy + 17 Medium + 5 Hard)

Core Problems:

- Climbing Stairs (Easy)
- House Robber (Medium)
- Coin Change (Medium)
- Longest Increasing Subsequence (Medium)
- Word Break (Medium)
- Unique Paths (Medium)
- Maximum Product Subarray (Medium)

Key Java Concepts: int[][] dp arrays, memoization with HashMap

Week 8: Advanced DP & Backtracking (25 problems)

Daily Target: 3-4 problems (2 Easy + 18 Medium + 5 Hard)

Core Problems:

- N-Queens (Hard)
- Permutations (Medium)
- Subsets (Medium)
- Combination Sum (Medium)
- Word Search (Medium)
- Edit Distance (Hard)
- Longest Common Subsequence (Medium)

Key Java Concepts: List<List<Integer>>, path management, backtracking with collections

Phase 3: Advanced Algorithms (Weeks 9-10)

Days 57-70 | Target: 50 More Problems

Week 9: Graphs & Union Find (25 problems)

Daily Target: 3-4 problems (2 Easy + 18 Medium + 5 Hard)

Core Problems:

- Course Schedule II (Medium)
- Number of Connected Components in an Undirected Graph (Medium)
- Graph Valid Tree (Medium)
- Accounts Merge (Medium)
- Word Ladder II (Hard)
- Alien Dictionary (Hard)

Key Java Concepts: Map<Integer, List<Integer>> adjacency lists, boolean[] visited arrays

Week 10: Advanced Topics & Heaps (25 problems)

Daily Target: 3-4 problems (2 Easy + 18 Medium + 5 Hard)

Core Problems:

- Kth Largest Element in an Array (Medium)
- Find Median from Data Stream (Hard)
- Implement Trie (Prefix Tree) (Medium)
- Word Search II (Hard)
- Design Twitter (Medium)
- LRU Cache (Medium)

Key Java Concepts: PriorityQueue<int[]>, custom Comparator, TrieNode implementation

Phase 4: Interview Preparation (Weeks 11-13)

Days 71-90 | Target: 90 More Problems

Week 11: Company-Specific Practice (30 problems)

Daily Target: 4-5 problems (1 Easy + 20 Medium + 9 Hard)

Focus Areas:

- Google: Graph algorithms, system design problems
- Amazon: Leadership principles problems, scalability
- Microsoft: Object-oriented design, recursion
- Meta: Social network problems, optimization
- Apple: Design problems, user experience focus

Week 12: Mock Interviews & Hard Problems (30 problems)

Daily Target: 4-5 problems (1 Easy + 20 Medium + 9 Hard)

Activities:

- Daily mock interviews with time constraints
- Practice explaining solutions clearly
- Focus on edge case handling
- Optimize for time and space complexity

Week 13: Final Sprint & Review (30 problems)

Daily Target: 4-5 problems (1 Easy + 20 Medium + 9 Hard)

Activities:

- Participate in daily LeetCode contests
- Review and strengthen weak areas
- Practice 3-4 problems in 90 minutes (interview simulation)
- Final comprehensive review of all patterns

Daily Intensive Schedule

Morning Block (8:00 AM - 12:00 PM) - 4 hours

- **8:00-8:30 AM:** Morning review and planning
- **8:30-10:00 AM:** Deep topic study (90 minutes focused learning)
- **10:00-10:15 AM:** Break
- **10:15-11:45 AM:** Problem solving session 1 (1-2 problems)
- **11:45-12:00 PM:** Break and lunch preparation

Afternoon Block (1:00 PM - 5:30 PM) - 4.5 hours

- **1:00-2:30 PM:** Problem solving session 2 (1-2 problems, different difficulty)
- **2:30-2:45 PM:** Break
- **2:45-4:15 PM:** Hard problem challenge (1 challenging problem)
- **4:15-4:30 PM:** Break
- **4:30-5:30 PM:** Review and analysis (mistakes, notes, progress tracking)

Evening Block (7:00 PM - 8:30 PM) - 1.5 hours

- **7:00-8:00 PM:** Mock interview practice (explain solutions aloud)
- **8:00-8:30 PM:** Final review and next day preparation

Problem Distribution Strategy

Core Categories (60% of problems)

1. **Arrays & Strings (40 problems):** Foundation for all coding interviews
2. **Dynamic Programming (30 problems):** Essential for optimization problems
3. **Trees & Binary Trees (25 problems):** Recursive thinking and traversals
4. **Company-Specific (25 problems):** Targeted preparation for dream companies

Pattern Categories (30% of problems)

5. **Hash Tables (25 problems):** Fast lookups and frequency counting
6. **Linked Lists (20 problems):** Pointer manipulation skills
7. **Graphs & DFS/BFS (40 problems):** Advanced traversal algorithms
8. **Two Pointers & Sliding Window (35 problems):** Optimization techniques

Specialized Topics (10% of problems)

9. **Advanced topics:** Heaps, Tries, Union-Find, Bit Manipulation
10. **System Design:** Object-oriented design problems

Java-Specific Optimization Guide

Essential Data Structure Templates

```
// ArrayList operations
List<Integer> list = new ArrayList<>();
list.add(element);
list.get(index);
list.remove(index);
```

```
// HashMap operations
Map<Integer, Integer> map = new HashMap<>();
map.put(key, value);
map.getOrDefault(key, defaultValue);
map.containsKey(key);

// PriorityQueue (Min Heap by default)
PriorityQueue<Integer> pq = new PriorityQueue<>();
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());

// Custom Comparator
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[0] - b[0]);

// StringBuilder for string operations
StringBuilder sb = new StringBuilder();
sb.append(char);
sb.toString();
```

Common Algorithm Templates

Two Pointers

```
int left = 0, right = arr.length - 1;
while (left < right) {
    if (condition) {
        left++;
    } else {
        right--;
    }
}
```

Sliding Window

```
int left = 0, maxlen = 0;
Map<Character, Integer> window = new HashMap<>();
for (int right = 0; right < s.length(); right++) {
    window.put(s.charAt(right), window.getOrDefault(s.charAt(right), 0) + 1);
    while (windowConditionViolated()) {
        // Shrink window
        left++;
    }
    maxlen = Math.max(maxlen, right - left + 1);
}
```

Binary Search

```
int left = 0, right = nums.length - 1;
while (left <= right) {
    int mid = left + (right - left) / 2;
    if (nums[mid] == target) {
        return mid;
    } else if (nums[mid] < target) {
        left = mid + 1;
    } else {
        right = mid - 1;
    }
}
```

Milestone Tracking System

Critical Checkpoints

- **Week 2 (Day 14):** 50 problems completed - Basic pattern recognition established
- **Week 4 (Day 28):** 100 problems completed - Comfortable with fundamental algorithms
- **Week 6 (Day 42):** 150 problems completed - Tackling medium-hard problems confidently
- **Week 8 (Day 56):** 200 problems completed - Advanced problem-solving skills developed
- **Week 10 (Day 70):** 250 problems completed - Company-specific preparation begins
- **Week 12 (Day 84):** 300 problems completed - Interview-ready proficiency achieved
- **Week 13 (Day 90):** 340+ problems completed - Contest-level performance

Assessment Criteria for Each Milestone

- **Pattern Recognition:** Can identify problem type within 2-3 minutes
- **Implementation Speed:** Complete Easy problems in 10-15 minutes
- **Code Quality:** Clean, readable, optimized solutions
- **Communication:** Explain approach clearly and handle follow-up questions
- **Edge Cases:** Identify and handle boundary conditions

Success Strategies

1. Consistency Over Intensity

- Solve 3-5 problems daily rather than cramming 20 problems in one day
- Maintain daily momentum rather than weekend bursts
- Track progress with a simple spreadsheet or app

2. Pattern-Based Learning

- Focus on understanding **why** a solution works, not just **how**
- Recognize that similar problems follow the same algorithmic patterns
- Create template solutions for each major pattern

3. Active Recall and Spaced Repetition

- Review problems after 3 days, 1 week, and 2 weeks
- Maintain a "retry list" for challenging problems
- Practice explaining solutions without looking at code

4. Mock Interview Practice

- Practice explaining your thought process aloud daily
- Use online platforms like Pramp for free mock interviews
- Record yourself solving problems to identify communication gaps

5. Company-Specific Preparation

- Research target companies' question patterns
- Use premium filters to access recent company questions
- Allocate 20% of practice time to company-specific problems

Common Pitfalls to Avoid

1. **Memorizing Solutions Instead of Patterns:** Focus on understanding the underlying algorithmic approach
2. **Skipping Easy Problems:** Build confidence and speed with fundamentals
3. **Not Practicing Under Time Pressure:** Use timers to simulate interview conditions
4. **Ignoring Edge Cases:** Always test boundary conditions and invalid inputs
5. **Poor Communication:** Practice verbalizing your thought process clearly
6. **Inconsistent Practice:** Daily practice beats weekend marathons
7. **Not Learning from Mistakes:** Analyze failed attempts to identify knowledge gaps

Week-by-Week Problem Lists

Week 1: Foundation - Arrays & Strings (25 problems)

1. Two Sum (Easy)
2. Remove Duplicates from Sorted Array (Easy)
3. Remove Element (Easy)

4. Search Insert Position (Easy)
5. Maximum Subarray (Easy)
6. Plus One (Easy)
7. Merge Sorted Array (Easy)
8. Pascal's Triangle (Easy)
9. Best Time to Buy and Sell Stock (Easy)
10. Best Time to Buy and Sell Stock II (Easy)
11. Single Number (Easy)
12. Majority Element (Easy)
13. Rotate Array (Medium)
14. Contains Duplicate (Easy)
15. Product of Array Except Self (Medium)
16. Missing Number (Easy)
17. Move Zeroes (Easy)
18. Intersection of Two Arrays II (Easy)
19. First Unique Character in a String (Easy)
20. Valid Anagram (Easy)
21. Valid Palindrome (Easy)
22. Reverse String (Easy)
23. Reverse Vowels of a String (Easy)
24. Implement strStr() (Easy)
25. Longest Common Prefix (Easy)

Week 2: Hash Tables & Two Pointers (25 problems)

1. Two Sum II - Input Array is Sorted (Easy)
2. 3Sum (Medium)
3. 3Sum Closest (Medium)
4. 4Sum (Medium)
5. Container With Most Water (Medium)
6. Trapping Rain Water (Hard)
7. Minimum Size Subarray Sum (Medium)
8. Subarray Product Less Than K (Medium)
9. Fruit Into Baskets (Medium)
10. Happy Number (Easy)
11. Count Primes (Easy)

12. Isomorphic Strings (Easy)
13. Word Pattern (Easy)
14. Intersection of Two Arrays (Easy)
15. Ransom Note (Easy)
16. Find the Difference (Easy)
17. Longest Palindrome (Easy)
18. Find All Anagrams in a String (Medium)
19. Group Anagrams (Medium)
20. Longest Consecutive Sequence (Medium)
21. Valid Sudoku (Medium)
22. Top K Frequent Elements (Medium)
23. Insert Delete GetRandom O(1) (Medium)
24. 4Sum II (Medium)
25. Subarray Sum Equals K (Medium)

Week 3: Sliding Window & Binary Search (25 problems)

1. Longest Substring Without Repeating Characters (Medium)
2. Minimum Window Substring (Hard)
3. Longest Substring with At Most Two Distinct Characters (Medium)
4. Longest Substring with At Most K Distinct Characters (Medium)
5. Longest Repeating Character Replacement (Medium)
6. Permutation in String (Medium)
7. Maximum Average Subarray I (Easy)
8. Max Consecutive Ones III (Medium)
9. Get Equal Substrings Within Budget (Medium)
10. Maximum Number of Vowels in a Substring (Medium)
11. Binary Search (Easy)
12. Search in Rotated Sorted Array (Medium)
13. Search in Rotated Sorted Array II (Medium)
14. Find Minimum in Rotated Sorted Array (Medium)
15. Find Minimum in Rotated Sorted Array II (Hard)
16. Find Peak Element (Medium)
17. Find First and Last Position of Element (Medium)
18. First Bad Version (Easy)
19. Guess Number Higher or Lower (Easy)

- 20. Sqrt(x) (Easy)
- 21. Valid Perfect Square (Easy)
- 22. Arranging Coins (Easy)
- 23. Heaters (Medium)
- 24. Find K Closest Elements (Medium)
- 25. Search a 2D Matrix (Medium)

Final Recommendations

Last Week Preparation Strategy (Days 85-90)

1. **Contest Simulation:** Participate in daily LeetCode contests
2. **Speed Practice:** Solve 3-4 problems in 90 minutes
3. **Communication Focus:** Practice explaining solutions clearly
4. **Company Review:** Focus on top 50 problems for target companies
5. **Behavioral Prep:** Prepare STAR format stories for behavioral interviews

Success Metrics

- **Technical Proficiency:** Solve 80%+ of Medium problems independently
- **Speed:** Complete Easy problems in 10-15 minutes, Medium in 20-30 minutes
- **Communication:** Explain approach and trade-offs clearly
- **Problem Recognition:** Identify patterns within first 2-3 minutes
- **Code Quality:** Write clean, optimized, production-ready code

This intensive 90-day program, when followed consistently with the prescribed daily schedule and problem progression, will provide comprehensive preparation for technical interviews at top technology companies. The key to success lies in maintaining consistent daily practice, focusing on pattern recognition over memorization, and developing both technical skills and communication abilities essential for interview success.